ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Low oxygen requires fast breathing.
2	**(b)** 6	$C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$.
3	**(a)** uses carbon dioxide and releases oxygen	Photosynthesis replenishes oxygen.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Valves prevent backflow.
5	**(a)** 1	$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$.
6	**(c)** chlorophyll is needed for photosynthesis	Chlorophyll.
7	**(c)** A is true but R is false.	Bile has no enzymes; it emulsifies fats.
8	**(c)** 4320	72 x 60.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Turgid guard cells bulge apart and open the pore; shrunken ones close it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; arteries carry blood under high pressure (hence thick walls) while valves are a separate feature of veins.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume per minute = 72 x 70 (1); = 5040 mL (5.04 L) (1).	1+1
12	Bile neutralises acid from the stomach (1); Pancreatic enzymes work in alkaline medium (1).	1+1
13	Low haemoglobin (1); Less oxygen to cells, less energy (1).	1+1
14	Reabsorbed 178.5 L (1); Percentage = 99.2% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Light is needed for photosynthesis (1); Starch is formed in light only (1); Covered region shows no starch (1).	1+1+1
16	Heart works continuously (1); Needs oxygen and glucose (1); Blockage leads to heart attack (1).	1+1+1
17	Salivary amylase breaks starch to maltose (1); Stomach acid stops amylase (1); Pancreatic amylase in small intestine gives glucose (1).	1+1+1
	OR: Atmospheric oxygen is lower (1); Faster breathing to get more oxygen (1); More RBC for transport (1).	1+1+1
18	Tubule reabsorption is faulty (1); Normally all glucose is reabsorbed (1); Diabetes may be a cause (1).	1+1+1
	OR: Soil solution is more dilute (1); Water moves into root hair cell (1); Moves into xylem (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Salivary amylase (1). (ii) Hydrochloric acid (1). (iii) Small intestine (1). (iv) Bile emulsifies fats, breaking large globules into smaller ones for the action of lipase (1); the pancreatic enzymes act only in an alkaline medium, so the acidic food from the stomach is made alkaline by bile (1).	1+1+1+2
20	Per minute: 7500 mL (1); Per hour: 450000 mL = 450 L (1); Oxygen = 21% of 450 = 94.5 L (1); Not all oxygen is absorbed (1); Exhaled air has about 16% oxygen (1).	1+1+1+1+1
	OR: Systolic 120: ventricle contracts (1); Diastolic 80: ventricle relaxes (1); Instrument: sphygmomanometer (1); Hypertension: high pressure strains arteries (1); May cause rupture or heart disease (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).

- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.
- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5040	70 x 72.
2	**(c)** more carbon dioxide	Exhaled air has about 4% CO_2.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Efficient oxygen supply.
4	**(c)** red blood cells	RBCs contain haemoglobin, which carries oxygen.
5	**(a)** reabsorption in nephron	Glucose is normally reabsorbed.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Short diffusion distance.
7	**(b)** 99%	(180 - 1.8)/180 = 99%.
8	**(d)** bacteria produce acid from sugar	Tooth decay.
9	**(c)** A is true but R is false.	Anaerobic breakdown is incomplete and releases much less energy than aerobic respiration; R is false.
10	**(d)** A is false but R is true.	Xylem carries water and minerals; food is carried by phloem, so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	NaOH absorbs CO_2 (1); No raw material for photosynthesis (1).	1+1
12	Phloem is cut (1); Food cannot reach the roots, roots die (1).	1+1
13	Produces food (glucose) for all organisms (1); Releases oxygen for respiration (1).	1+1
14	No gas exchange (1); No photosynthesis or transpiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Herbivores: long for cellulose digestion (1); Carnivores: short (1); Meat is easier to digest (1).	1+1+1
16	Evaporation increases (1); Wind removes humid air (1); Cooling and water absorption (1).	1+1+1
17	Green plant: autotrophic (1); Cuscuta: parasitic, lacks chlorophyll (1); Absorbs food from host (1).	1+1+1
	OR: Valves in heart and veins (1); Muscular contractions of heart (1); Pressure gradient arteries to veins (1).	1+1+1
18	Photosynthesis stores energy; respiration releases (1); Photosynthesis uses CO_2; respiration gives CO_2 (1); Photosynthesis only in light; respiration all the time (1).	1+1+1
	OR: Glomerulus has fine capillaries (1); High pressure filtration (1); Long tubule for reabsorption (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Four (1). (ii) Pulmonary vein (1). (iii) Sphygmomanometer (1). (iv) Birds and mammals need a high supply of oxygen as they have high energy needs and maintain constant body temperature (1); separation allows a highly efficient supply of oxygen to the body (1).	1+1+1+2
20	Fish: two-chambered heart (1); Single circulation (1); Human: four chambers, double circulation (1); Warm-blooded need more oxygen (1); Prevents mixing (1).	1+1+1+1+1
	OR: Heart rate rises (1); Cardiac output rises (1); Breathing rate rises (1); Muscles may respire anaerobically (1); Lactic acid accumulates if oxygen supply is inadequate (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).

- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** phloem	Phloem is outside of the xylem.
2	**(a)** Both A and R are true and R is the correct explanation of A.	ATP drives loading.
3	**(b)** ethanol and carbon dioxide	Fermentation gives ethanol and CO2.
4	**(d)** away from the heart under high pressure	Arteries are away from heart.
5	**(c)** A is true but R is false.	A is true. R is false: guard cells take up water in light and swell.
6	**(c)** cytoplasm	Glycolysis occurs in the cytoplasm.
7	**(a)** urea	Urea and salts.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Replaces kidney function.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Villi (rich in blood vessels) greatly increase the absorbing surface.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but the second statement describes a different part of the excretory system.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$C_6H_{12}O_6 \longrightarrow 2\,C_2H_5OH + 2\,CO_2 + \text{energy}$ (2).	2
12	It brings oxygenated blood (1); Pulmonary vein is the only vein carrying oxygenated blood (1).	1+1
13	Higher pressure (1); Oxygen supply at depth is limited (1).	1+1
14	Pulmonary: heart to lungs and back (1); Systemic: heart to body and back (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inhalation (1); Diaphragm flattens to increase chest volume (1); Air cannot enter properly (1).	1+1+1
16	Breathing rate increases (1); Heart rate increases (1); More blood to the muscles (1).	1+1+1
17	Contraction raises ribs and flattens diaphragm (1); Chest volume increases, pressure falls (1); Air rushes in; reverse expels (1).	1+1+1
	OR: Alveoli bring oxygen near blood (1); Haemoglobin binds oxygen (1); Released in oxygen-poor tissues (1).	1+1+1
18	Urea and wastes accumulate in the blood (1); Salt and water balance is lost (1); Poisoning leads to death (1).	1+1+1
	OR: Fish have low energy needs (1); Single circulation: blood passes heart once (1); Mammals need high pressure and oxygen (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Nephron (1). (ii) Ureter (1). (iii) Urea (1). (iv) So that nitrogenous waste diffuses from the blood into the fluid along the concentration gradient (1); equal osmotic pressure prevents gain or loss of water by the blood (1).	1+1+1+2
20	No digestion in mouth (1); Bile emulsifies fat in the intestine (1); Lipase breaks to fatty acids and glycerol (1); Absorbed by villi (1); Taken by lymph (1).	1+1+1+1+1
	OR: Urea builds up (1); Water and salt imbalance (1); Dialysis filters blood (1); Needs repeated sessions (1); Transplant is the permanent cure (1).	1+1+1+1+1

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